

LAB 4

Storage and Query Benchmark

Healthcare Admissions Dataset · ~55,500 rows · pandas vs PostgreSQL vs DuckDB

Query Benchmarked

Average billing amount and admission count, grouped by medical condition run identically across three tools: pandas on CSV, PostgreSQL, and DuckDB on Parquet.

File Size Comparison

Format	Size	Reduction
CSV	8,202.4 KB	—
Parquet	2,755.6 KB	~3.0x smaller

Query Time Comparison

Tool	Cold Run	Warm (Repeated) Run
pandas (CSV)	0.0192 s	0.0449 s
PostgreSQL	0.0277 s	0.0297 s
DuckDB (Parquet)	0.2636 s	0.0093 s ★ fastest

KEY FINDING

DuckDB's cold run was the slowest of the three, contrary to the unit's illustrative claim. The 0.2636s cold-run cost came from one-time engine initialization and first-time Parquet file access not the query itself. Rerunning the identical query dropped to 0.0093s, faster than pandas or PostgreSQL, while both of those stayed roughly stable across repeated runs. This is a direct, self-measured example of the unit's warning against "believing benchmarks you never ran."

Verdict

We choose DuckDB with Parquet. Once warmed up, it was the fastest of all three tools tested, and its Parquet file is roughly 3x smaller than the raw CSV. Its one-time cold-start cost is a session-level engine initialization, not a repeated query cost, so it does not change the recommendation. DuckDB gives Postgres-level reliability with pandas-level simplicity and zero server setup the right tool for this dataset now, and it scales better if the data grows later.